cannot be neglected in comparison with the externally applied field. In paramagnetic materials where the crystal field is negligible, Eq. (8.48) holds good. But when it is strong, it may break the rotational symmetry of the dipole and affect its total angular momentum. Also, the average value of $L_z$ may reduce to zero which is known as quenching of the orbital angular momentum. In such a case, Eq. (8.39) should be written as

$$E = 2m_s\mu_B B \qquad [\text{As } g = 2]$$

$$= \left(\frac{e\hbar}{m}\right) m_s B \tag{8.50}$$

For $\left(\frac{e\hbar}{m}\right) m_s B << kT$, it yields

$$\chi = \frac{\mu_0 N}{3kT}\left(\frac{e\hbar}{m}\right)^2 S(S+1) \tag{8.51}$$

For example, for $Mn^{3+}$ ions, the value of $\chi$ as obtained from the above expression with $S = 2$ is in conformity with the experimental observations where Eq. (8.48) predicts a zero susceptibility.

## 8.5 FERROMAGNETISM

Like paramagnetism, ferromagnetism is also associated with the presence of permanent magnetic dipoles, but unlike paramagnetism, the magnetic moments of adjacent atoms in this case are aligned in a particular direction even in the absence of the applied magnetic field. Thus a ferromagnetic material exhibits a magnetic moment in the absence of a magnetic field. The magnetization existing in a ferromagnetic material in the absence of an applied magnetic field is called the *spontaneous magnetization.* It exists below a certain critical temperature called the *Curie temperature,* $T_C$, The alignment of magnetic moments below the Curie temperature is due to the exchange interaction between the magnetic ions and will be described later. Above the Curie temperature, the thermal effects offset the spin alignment and the ferromagnetic substance becomes paramagnetic.

The ferromagnetic substances acquire a large magnetization in the presence of even a weak external magnetic field. They possess a large and positive value of susceptibility which, in general, is not constant but varies with field strength. The variation of magnetization with field strength exhibits the well-known hysteresis curve. The examples of ferromagnetic materials are the elements such as Fe, Ni, Co, Gd and Dy, and a number of alloys and oxides such as MnBi, MnAs, $CrO_2$, etc.

### 8.5.1 Weiss Theory of Ferromagnetism

The theory of ferromagnetism put forward by Weiss is centred about the following two hypotheses :

(i) A specimen of ferromagnetic material contains a number of small regions called *domains* which are spontaneously magnetized. The magnitude of spontaneous magnetization of the specimen as a whole is determined by the vector sum of the magnetic moments of individual domains.

(ii) The spontaneous magnetization of each domain is due to the presence of an exchange field, $B_E$, which tends to produce a parallel alignment of the atomic dipoles. The field $B_E$ is assumed to be proportional to the magnetization $M$ of each domain, i.e.,

$$B_E = \lambda M \tag{8.52}$$

where $\lambda$ is a constant called the *Weiss-field constant* and is independent of temperature.

It may be noted that, in the above expression, we always use the average values of $B_E$ and $M$ for a domain. The field $B_E$ is also called the *molecular field* or the *Weiss field* and is generally quite strong as compared to the applied field $B$. For iron, for example, $B_E \approx 1000\ T$. Thus the effective mangetic field on an atom or ion becomes

$$B_{eff} = B + B_E = B + \lambda M \tag{8.53}$$

In order to develop the theory of ferromagnetism, we shall use the quantum theory of magnetization rather than the classical Langevin theory used by Weiss originally.

Consider a ferromagnetic solid containing $N$ atoms per unit volume each having a total angular momentum quantum number $J$. By analogy to Eq. (8.46), we can write the expression for magnetization as

$$M = NgJ\mu_B B_J(x) \tag{8.54}$$

where

$$B_J(x) = \frac{2J+1}{2J}\coth\left(\frac{2J+1}{2J}\right)x - \frac{1}{2J}\coth\left(\frac{x}{2J}\right) \tag{8.55}$$

and

$$x = \frac{gJ\mu_B B_{eff}}{kT} = \frac{gJ\mu_B}{kT}(B + \lambda M) \tag{8.56}$$

In case of spontaneous magnetization, $B = 0$ and Eq. (8.56) becomes

$$x = \frac{gJ\mu_B\lambda M}{kT} \tag{8.57}$$

$$\therefore \quad M(T) = \frac{xkT}{\lambda g J \mu_B} \tag{8.58}$$

As $T \to 0$ or $x \to \infty$, $B_J(x) \to 1$; the magnetic moments align themselves parallel to the field and the magnetization $M$ becomes the *saturation magnetization*, $M_s(0)$. Thus, from Eq. (8.59) give

$$M_s(0) = NgJ\mu_B \tag{8.59}$$

From Eqs. (8.58) and (8.59), we obtain

$$\frac{M(T)}{M_s(0)} = \frac{xkT}{\lambda N g^2 J^2 \mu_B^2} \tag{8.60}$$

Also, Eqs. (8.54) and (8.58) give

$$\frac{M(T)}{M_s(0)} = B_J(x) \tag{8.61}$$

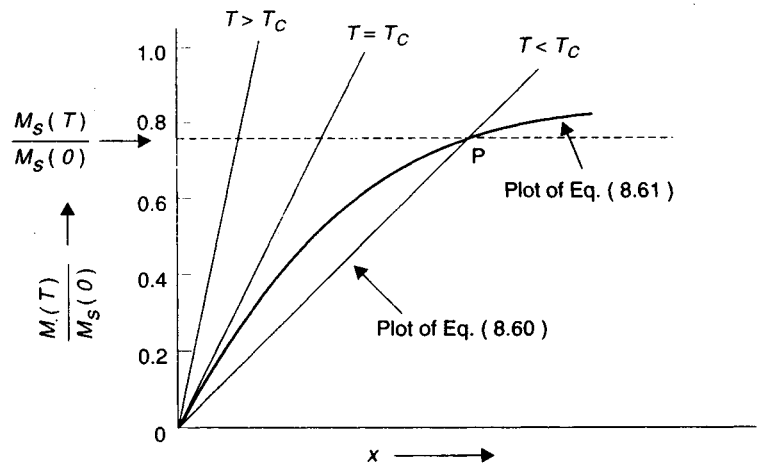



Fig. 8.3. *Graphical solution of the simultaneous equations (8.60) and (8.61). A point of intersection P determines the spontaneous magnetization $M_s(T)$ at a given temperature. No solution exists for $T > T_C$.*

The magnetization $M(T)$ at a given temperature can be obtained by solving Eqs. (8.60) and (8.61) simultaneously. The plots of $M(T)/M_s(0)$ versus

$x$ representing Eqs. (8.60) and (8.61) are shown in Fig. 8.3. Note that the Eq. (8.60) represents a straight line passing through the origin and having slope proportional to $T$. At temperature equal to the critical temperature $T_C$, this line is tangent to the Brillouin function at the origin. The intersection of the two plots at the point O represents a positive solution, but the magnetization corresponding to this point is unstable. Another point of intersection appears for $T < T_C$ which indicates a non-zero value of $M$ even for zero external field and hence corresponds to spontaneous magnetization. It also follows that the spontaneous magnetization decreases with increase in temperature and vanishes beyond the temperature $T_C$ which is known as the *ferromagnetic Curie temperature*. The variation of spontaneous magnetization with temperature is shown in Fig. 8.4. It becomes maximum at 0 K when all the moments are lined up in a particular direction under the influence of the exchange field. This variation of $M_s$ enables one to classify the paramagnetic-ferromagnetic transition as a second order phase transition, i.e., a transition characterized by an order parameter (e.g. $M_s$ in this case) which is non-zero only below $T_C$.

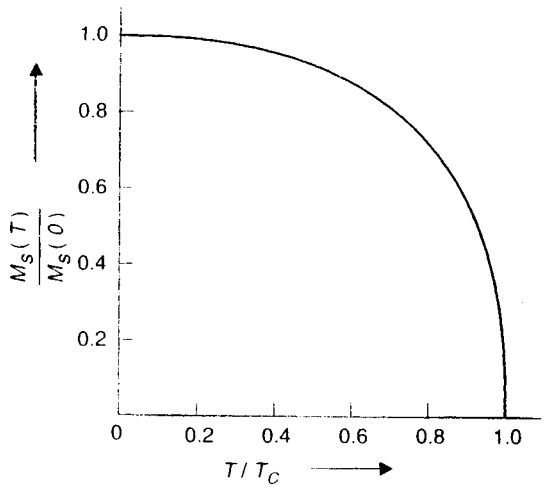



Fig. 8.4. *Spontaneous magnetization versus temperature for* $T < T_C$.

**The Paramagnetic Region**

Consider the magnetization in the region well above the Curie temperature. For $T > T_C$, the spontaneous magnetization is zero and an external field will have to be applied to produce some magnetization. This field should, however, be weak enough to avoid the saturation state. In such a state, we find from Eq. (8.56) that $x << 1$ and we can write

$$B_J(x) \cong \left(\frac{J+1}{3J}\right)x$$

Therefore, the expression (8.54) becomes

$$M = Ng\mu_B (J+1)x/3$$

$$x = \frac{gJ\mu_B(B+\lambda M)}{kT} \tag{8.62}$$

Thus,

$$M = \frac{Ng^2\mu_B^2 J(J+1)}{3kT}(B+\lambda M) \tag{8.63}$$

which gives

$$\chi = \frac{\mu_0 M}{B} = \frac{\mu_0 T_C/\lambda}{T-T_C} = \frac{C}{T-T_C} \tag{8.64}$$

where

$$C = \frac{\mu_0 T_C}{\lambda} \tag{8.65}$$

and

$$T_C = \frac{\lambda N g^2 \mu_B^2 J(J+1)}{3k} \tag{8.66}$$

The expression (8.64) is known as the *Curie-Weiss law.* It satisfactorily describes the temperature dependence of susceptibility in the paramagnetic region provided the temperature is well above the Curie temperature. The Curie temperature determined from the theory of spontaneous magnetization differs by a few degrees from the experimentally determined value for the paramagnetic region.

### 8.5.2 Nature and Origin of Weiss Molecular Field : Exchange Interactions

The Weiss theory of ferromagnetism is based on the concept of ferromagnetic domains which are spontaneously magnetized due to the presence of an internal molecular field called the *Weiss field* or the *exchange field,* $B_E$. The theory, however, does not explain the origin and nature of this field. The Weiss field cannot be simply due to magnetic dipole-dipole interaction between the neighbouring dipoles as this would generate fields of the order of $10^3$G only whereas the actual field strengths are observed to be quite high. For example, the Weiss field for iron is of the order of $10^7$ G. It was Heisenberg who first proposed in 1928 that the Weiss field was the consequence of the quantum-mechanical exchange interaction between the atoms. This interaction arises due to the Pauli's exclusion principle according to which any change in the relative orientation of the two spins would disturb

the spatial distribution of charge, thus producing interaction between the two atoms. Apparently, the strength of exchange interaction between the adjacent atoms depends on the extent of overlap of their wave functions as well as the relative orientation of the electron spins but not on the spin magnetic moments. Thus this is an electrostatic and non-magnetic type of interaction.

Using Heitler-London theory of chemical bonding, it can be shown that the total energy of a system of two atoms contains an exchange energy term given by

$$U_{ij} = -2J_e \mathbf{S_i}.\mathbf{S_J} \tag{8.67}$$

where $\mathbf{S_i}$ and $\mathbf{S_j}$ represent the spins of the two atoms and $J_e$ is the exchange integral which is assumed to be the same for any pair of atoms. Its value depends on the overlap of the charge distributions of the two atoms, i.e., on the interatomic distance. In general, $J_e$ is positive for large interatomic distances and negative for smaller ones. The expression (8.67) is known as the *Heisenberg model* of exchange energy. It also follows from (8.67) that if $J_e$ is positive, the parallel arrangement of spins exhibits lower energy and hence is more stable as compared to the antiparallel arrangement, thereby producing magnetization. In a similar way, it can be concluded that the negative value of $J_e$ does not favour magnetism.

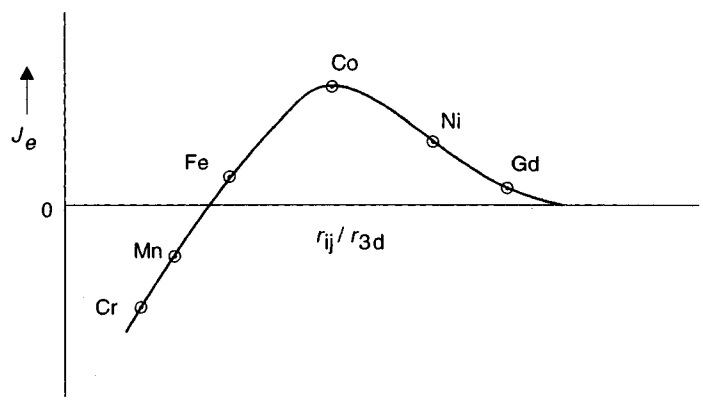



Fig. 8.5. *Plot of exchange integral, $J_e$ versus ratio of interatomic separation to the radius of the 3d orbit, $r_{ij}/r_{3d}$*

The exchange integral for iron group of atoms is found to be positive inspite of the fact that the interatomic distance for this group is not large. This follows from the observations of Bathe and Slater which indicate that $J_e$ is positive if the ratio $r_{ij}/r_{3d}$ is greater than 3 but is not much larger than 3, $r_{ij}$ being the distance between the atoms $i$ and $j$ and $r_{3d}$ is the radius of the unfilled

3*d*-shell. The variation of $J_e$ as a function of $r_{ij}/r_{3d}$ is shown in Fig. 8.5. It is apparent that the ratio $r_{ij}/r_{3d}$ favours ferromagnetism for Fe, Co, Ni and Gd whereas for Mn and Cr it does not favour ferromagnetism. Thus elements like Mn and Cr do not exhibit ferromagnetism.

The expression for the exchange field, $B_E$, and the relationship between $J_e$ and $\chi$ can be obtained by following the procedure suggested by Stoner. Assuming $J_e$ to be constant for all the neighbouring pairs and considering contributions from the nearest neighbours only, we may write the exchange energy of an $i$th atom as

$$U_i = -2J_e \sum_j \mathbf{S_i}.\mathbf{S_j} \tag{8.68}$$

where the summation is over the nearest neighbours of the $i$th atom. Stoner replaced the instantaneous values of the neighbouring spins by their time averages. For $z$ nearest neighbours, (8.68) can be written as

$$U_i = -2zJ_e\,(S_{xi}\langle S_{xj}\rangle + S_{yi}\langle S_{yj}\rangle + S_{zi}\langle S_{zj}\rangle) \tag{8.69}$$

For magnetization, $M$, along the $z$-axis, we have

$$\langle S_{xj}\rangle = \langle S_{yj}\rangle = 0 \text{ and } \langle S_{zj}\rangle = \frac{M}{g\mu_B N} \tag{8.70}$$

$$\therefore \qquad U_i = -\frac{2zJ_e S_{zi} M}{gN\mu_B} \tag{8.71}$$

It may be written as

$$U_i = -gS_{zi}\mu_B B_E \tag{8.72}$$

with

$$B_E = \frac{2zJ_e}{Ng^2\mu_B^2} M \tag{8.73}$$

The factor $\dfrac{2zJ_e}{Ng^2\mu_B^2}$ is a constant and is equal to $\lambda$, i.e.,

$$\lambda = \frac{2zJ_e}{Ng^2\mu_B^2} \tag{8.74}$$

and Eq. (8.73) gives

$$B_E = \lambda M$$

which is the same as (8.52).

### 8.5.3 Concept of Domains and Hysteresis

According to the Weiss theory described above, the exchange interactions between the neighbouring dipoles in a ferromagnetic material generate an internal exchange field $B_E$ which aligns them in a particular direction. It is, however, observed that a ferromagnetic material such as iron does not exhibit a net magnetization unless it is placed in an external field. Weiss explained this by introducing the concept of ferromagnetic domains. According to this concept, a single crystal of a ferromagnetic solid is divided into a number of small regions called *domains* each one of which is spontaneously magnetized by the exchange field. The magnetization vectors of different domains are, however, randomly oriented so that no net magnetization is produced in the material as a whole. In the presence of an external magnetic field, the domains pointing in the direction of the field grow at the expense of those pointing in other directions, thereby resulting in some non-zero magnetization in the material.

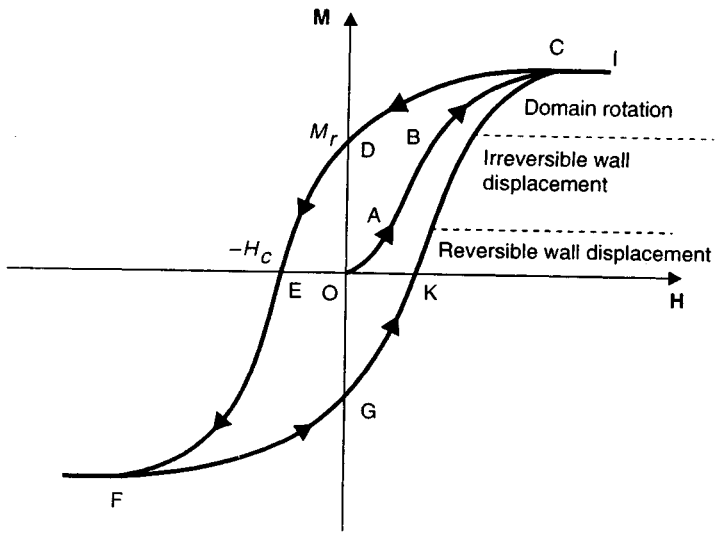



Fig. 8.6. *Typical hysteresis curve for a ferromagnetic material indicating the predominant magnetization processes in different regions.*

According to Neel, the origin of domain structure of a ferromagnetic solid rests in the principle of minimization of the total energy of the material which, apart from exchange energy, comprises the magnetic field energy, anisotropy energy, domain wall energy and magnetostrictive exchange energy. As an example, the presence of free magnetic poles at the ends of a magnetized specimen generates an external field **H** which gives rise to the magnetic field energy equal to ½. **H.B**. This energy can be lowered by

reducing the volume of the domain by subdividing it into neighbouring subdomains, thereby reducing the field **H** and hence the magnetic field energy. However, such a subdivision of domains cannot continue indefinitely and is restrained by the fact that the formation of domain walls requires additional energy. Thus, it can be concluded that, whereas the presence of large domains is energetically unfavourable, a ferromagnetic material must possess a domain structure consisting of a number of smaller domains which corresponds to a state of minimum energy.

All ferromagnetic materials exhibit the well known hysteresis curves; a typical one is shown in Fig. 8.6. It is apparent that for $T<T_C$, there are two solutions for $M$ (and hence for $B$) which trace the boundary of the hysteresis curve. These solutions can be obtained from the intersection of the plots of Eq. (8.61) and the expression

$$M = \left(\frac{kT}{\mu_B g \lambda J}\right) x - \frac{1}{\lambda} B \tag{8.75}$$

where the latter expression has been obtained from Eq. (8.62). The physical cause of the phenomenon of hysteresis can be understood from the concept of domains. The magnetization produced in a ferromagnetic solid in the presence of an external magnetic field may be attributed to

(a) growth in the size of domains having favourable orientation with respect to the applied field at the expense of those having unfavourable orientation, and

(b) rotation of the directions of magnetization of various domains along the field direction.

These two magnetization processes are illustrated below :

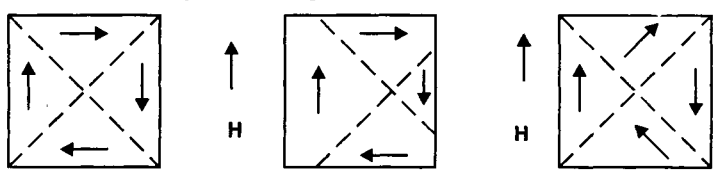



Fig. 8.7. *The two fundamental processes of magnetization.*

When a small magnetic field is applied across a ferromagnetic material, the domains pointing almost along the direction of the field grow at the expense of the domains having opposite orientation, thus resulting in a small magnetization as indicated by the initial portion (OA) of the hysteresis curve. Such displacements of domain boundaries are mostly reversible and hence